

Synopsis

The insurance broking industry continues to face operational challenges due to fragmented data systems, manual workflows, and inefficient commission management practices. These issues often lead to data inconsistencies, processing delays, lack of transparency, and difficulty in decision-making. To address these challenges, this project presents a centralized web-based application titled **Leads and Report Analytics using Machine Learning on Insurance Sales Data**, designed specifically for insurance broking companies to streamline and automate their core operations.

The proposed system integrates lead and customer management, policy and endorsement handling, payment and commission tracking, document storage, and real-time reporting into a single secure digital platform. The system is built using **FlutterFlow** for a responsive cross-platform user interface, **Supabase** for backend services such as authentication, real-time updates, and storage, and **PostgreSQL** as the centralized relational database. Machine learning models are implemented using Python libraries such as Scikit-learn, ensuring accuracy and consistency in predictive analytics.

A key feature of the system is the integration of two supervised machine learning models. The **Predictive Renewal Churn Model** analyzes historical data such as payment behavior, policy tenure, and customer attributes to identify customers at risk of policy lapse, assigning risk scores categorized as High, Medium, or Low. The **Lead Conversion Scoring Model** evaluates attributes such as customer profile, expected premium, and interaction history to assign a conversion probability score, categorizing leads into Hot, Warm, Cool, and Cold segments. These models enable brokers to make proactive decisions, prioritize high-value leads, and improve customer retention strategies.

The system architecture follows a three-tier structure consisting of a presentation layer (FlutterFlow frontend), an application layer (Supabase backend), and a data layer (PostgreSQL database). The platform supports role-based access control, multi-agent management, and cloud-based document handling, enabling secure and efficient operations. Interactive dashboards and analytical reports provide real-time insights into key performance indicators such as active policies, premium amounts, pending renewals, lead conversion rates, and commission summaries.

Compared to existing platforms used by insurance companies such as LIC, HDFC Life, ICICI Prudential, and SBI Life, the proposed system goes beyond transactional processing to offer real-time predictive analytics integrated directly into the user workflow. Unlike traditional systems where analytical tools and CRM functionalities operate in silos, this platform provides a unified environment where machine learning insights are accessible to brokers and sales agents in real time, enabling data-driven decision-making at the operational level.

The project demonstrates how a modern, cloud-enabled architecture can transform traditional insurance broking processes into a reliable, efficient, and scalable digital ecosystem. By combining structured database management with machine learning and intuitive user interface design, the system significantly reduces manual effort, minimizes errors, and enhances decision-making capabilities. The system is developed with synthetic data for training and testing purposes and has been validated through iterative testing, achieving a cross-validated AUC of 0.86 on the churn prediction model. Overall, this solution serves as a practical and impactful contribution to the evolving needs of the insurance broking industry.